Root-vegetable purées (single-ingredient) — HMTc certification standard

Institute of Contaminant Standards — Heavy Metal Tested & Certified program

Version 1.0 · Native basis: as_consumed · 7/8 analytes with a published limit

STANDARD

root-vegetable-purees-single-ingredient, version 1.0. HMTc Method v2.0 (government-limit default, 8:1 basis, read-across)

PROVENANCE

Compiled from data/evidence/regulatory_limits.csv. Each limit is a lookup against sovereign law, not an occurrence computation.

TAMPER-EVIDENT SEAL

The SHA-256 of this exact PDF is sealed by an RFC-3161 trusted timestamp. The token (standard.pdf.tsr) and the manifest (manifest.json) accompany this file; the date of record is the timestamping authority's, not the program's.

Abstract

This is the frozen, version-pinned Heavy Metal Tested & Certified standard for root-vegetable purées (single-ingredient). Each per-analyte certification limit is the strictest heavy-metal maximum level set by any government in the world for this product class, expressed in the product's native basis (as_consumed) and derived by lookup against sovereign law under HMTc Method v2.0 (government-limit default, 8:1 basis, read-across). The per-jurisdiction derivation below is the complete "show your work". This snapshot is immutable: any material change to a limit mints a new version with its own timestamped snapshot.

1 | Certification limits

All values in ppb (µg/kg), as_consumed basis. A brand meeting every published value on a like-for-like basis qualifies for the mark. Tier 1 toxics (Pb, Cd, iAs, mercury) carry a zero-tolerance posture; Tier 2 metals (Ni, Al, Sn, total Cr) carry the program's transitional allowance. Where a limit is set in another basis by the binding jurisdiction it is converted into the native basis and the conversion is named in the "Basis derivation" column.

both

ANALYTE	TIER	LIMIT (PPB)	BINDING JURISDICTION	BASIS DERIVATION	RATIONALE
Lead (Pb)	Tier 1	20	European Union	face value	regulatory-alignment
Inorganic arsenic (iAs)	Tier 1	20	European Union	face value	regulatory-alignment
Cadmium (Cd)	Tier 1	40	European Union	face value	regulatory-alignment

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Mercury (tHg)	Tier 1	8	—	face value (nearest government value; comparator vs any direct limit)	read_across
Nickel (Ni)	Tier 2	500	European Union	face value	regulatory-alignment
Aluminium (Al)	Tier 2	500	—	face value (nearest government value; comparator vs any direct limit)	read_across
Tin (Sn)	Tier 2	5000	—	face value (nearest government value; comparator vs any direct limit)	read_across
Total chromium (Cr)	Tier 2	monitored	—	—	not_material

2 | Per-jurisdiction derivation

The binding value for each analyte is the lowest sovereign maximum level across the jurisdictions on record, in the native basis. Rows marked "converted" were set by the jurisdiction in the other basis and carried across the reconstitution ratio; rows marked "native" were already in this basis. Every value is keyed to its record in the sovereign register.

both

ANALYTE	JURISDICTION	LIMIT (PPB)	BASIS	SOVEREIGN REGISTER ID
Pb	European Union-(binding)	20	converted	eu2023_pb_babyfood_cereal_20
—	US-FDA	20	converted	fda2025_pb_root_baby_food_20
—	Gulf (GSO)	100	converted	gso193_2021_pb_root_and_tuber_vegetables_-100
—	HK	100	converted	hk_cap132v_pb_root-and-tuber-vegetables_100
—	India	100	converted	in_fssai2016_pb_014_root_and_tuber_-vegetables_100
—	PH	100	converted	ph_pns194_2017_root_tuber_vegetables_pb_100
—	Thailand	100	converted	th_moph414_2020_pb_root_tuber_vegetables_-100
—	Singapore	200	converted	sg_foodregs_2026_baby_food_pb_200
iAs	European Union-(binding)	20	converted	eu2023_ias_babyfood_20
Cd	European Union-(binding)	40	converted	eu2023_cd_babyfood_cereal_40
—	Gulf (GSO)	40	converted	gso193_2021_cd_processed_cereal_based_-foods_and_baby_foods_for_infants_and_-young_childr_40

—	United Kingdom	40	converted	uk_cd_processed-cereal-based-foods-and--baby-fo_40
—	Australia / NZ	100	converted	aunz_sch19_cd_root_tuber_100
—	HK	100	converted	hk_cap132v_cd_root-and-tuber-vegetables_100
—	PH	100	converted	ph_pns194_2017_root_tuber_vegetables_cd_100
Ni	European Union-(binding)	500	converted	eu2024_ni_babyfood_500

3 | Method and provenance

HMTc Method v2.0 (government-limit default, 8:1 basis, read-across). The default standard for any (product, analyte) is the strictest maximum level set by a credible government regulator, converted to the row's native basis. Where the law itself splits a limit by formulation, the standard adopts the law's own split and cites it. Species limits are bounded by their parent total (iAs ≤ tAs, MeHg ≤ tHg, Cr-VI ≤ total Cr); read-across limits (for example aluminium carried from an as-consumed limit to an as-sold basis across the reconstitution ratio) are disclosed as read-across and never presented as direct governance.

The occurrence and toxicology corpus that contextualises these limits is maintained independently as the Heavy Metal Index literature baseline (heavymetalindex.com). This certification standard cites that baseline one way; it does not use the baseline's contents as the basis for any government-limit value stated here.

4 | Version and tamper-evidence

This is version 1.0 of the Root-vegetable purées (single-ingredient) standard. A certification number becomes final when its version carries a resolvable DataCite DOI; a material change to any limit mints a new version with its own frozen snapshot, so a superseded number is never silently overwritten. Tamper-evidence does not depend on trusting the program: the SHA-256 of this exact PDF was submitted to an RFC-3161 trusted timestamping authority (https://freetsa.org/tsr), which returned a signed token binding that hash to a time it — not the program — attests. The token (standard.pdf.tsr), the authority's CA chain (tsa-cacert.pem), and the manifest (manifest.json) travel with this file.

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